A REPRODUCIBLE WORKFLOW FOR LLM-PROPOSED MATHEMATICAL RESULTS: ERDŐS-TEMPLATE PROMPTS, ADVERSARIAL VERIFICATION SUBAGENTS, AUTOMATED LITERATURE DIFF, AND ACCUMULATING CALIBRATION

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Abstract. Frontier LLMs (ChatGPT-Pro, Claude Opus 4.7, etc.) can propose non-trivial mathematical results—closed forms, identity verifications, structural arguments—at a rate substantially higher than any individual researcher. The bottleneck is no longer *generation* but *filtering*: separating genuine novel contributions from rediscoveries of classical material, from plausible-looking but incorrect derivations, and from calibrated-honest "I don't know" responses that the model already correctly produces. We document a four-component workflow developed during a 30-cycle private research run (Ramanujan-mining project, 2026-05) that filters LLM proposals reliably enough to produce publication-grade output: (1) an Erdős-style prompt template with open T/F formulation, machinery loading, anti-handwave guards, and escape hatches; (2) domain-specific machinery loading that accumulates context across cycles; (3) a verification gate of adversarial subagents (independent reimplementation + numerical verification + literature diff); and (4) diary-discipline + a "did not pan out" log that compounds calibration over time. We report the cycle-by-cycle hit rate: of approximately 30 substantive ChatGPT-Pro cycles, ~ 7 produced calibrated-honest negatives that the model correctly flagged ("I do not have a closed form for this"), ~ 16 produced confirmed-known content (useful as implementations, not as novel-math claims), and 1 produced a genuinely novel mathematical finding (a correction to a published Andrews–Berndt commentary on Ramanujan's *Lost Notebook* Entry 4.2.3). The methodology's strongest single output is the *verification-gate pattern*: 4 of 4 adversarial subagents returned actionable verdicts that either confirmed, partially confirmed, or downgraded a prior LLM claim; in each case the verdict was the one that survived in the final paper output. We release the full workflow, prompt templates, verification scripts, and diary record at https://github.com/jounih/ramanujan-mining for reproduction and critique.

1. Introduction

A frontier LLM can be a remarkably productive mathematical collaborator. Given a well-posed problem in a specific subfield, it can frequently propose a closed-form expression, an identity, or a structural argument that would take a human researcher hours to construct from scratch. But the same LLM, given the same problem, will sometimes also propose a plausible-looking but incorrect answer; or rediscover a classical result without flagging the prior art; or overclaim "novel" status for something derivable in five lines of textbook calculation; or—and this is the most useful case—honestly report that it cannot solve the problem and identify the specific obstruction.

The bottleneck for productive use of LLM mathematical proposers is therefore not generation but filtering. This paper reports a workflow we developed during a private 30-cycle research run (Ramanujan-mining, May 2026) that filters LLM proposals reliably enough to produce publication-grade output: one short corrigendum (Andrews–Berndt *Lost Notebook* Part IV, Entry 4.2.3) and five verified computational primitives covering ML attention, numerical PDEs, port-Hamiltonian simulation, asymptotic special-function evaluation, and Madelung lattice sums.

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We argue that the workflow generalizes beyond Ramanujan-style work, and we document it here for reproduction.

2. The four-component workflow

2.1. **Erdős-style prompt template.** Every productive prompt in the run followed roughly the same skeleton, modeled after OpenAI's chain-of-thought demonstrations on Erdős-style problems [Ope26]:

```
CONTEXT. [1-3 paragraphs loading the specific machinery: operator
basis, recent verified results, catalogues to check against,
recent gotchas]

OPEN QUESTION. [The actual T/F or construct-or-prove-impossible
deliverable]

PRIORS / GUARDRAILS. [What most experts expect; what the failure
modes look like; specific things to avoid]

DELIVERABLE FORMAT. [Concrete output schema -- explicit LaTeX +
numerical check at N digits + cost ratio + provenance]

IF YOU CAN'T. [Escape hatch -- "tell me the obstruction" -- turns
failure into information]
```

Four design choices in this template matter:
Open T/F formulation. "Does X hold, or doesn't it?" rather than "Prove X" or "Disprove X". The first lets the model pick the productive direction; the second two bias toward one answer.
Specific numerical bar. "≥ 30 digits agreement," "$\leq 10^{-100}$ relative error," "$\geq 3\times$ wall-clock speedup." Vague verification gates produce vague answers.
Anti-handwave clause. " 'It's likely hard' is not an answer." Pushes the model to either commit to a construction or commit to a precise structural argument.
Escape hatch. "If you can't, tell me the obstruction." Converts failure into information. The 7 calibrated-honest cycles in our run (see section 4) all followed this pattern—the model returned a clean "no" with the specific structural obstruction identified, rather than a fabricated answer.

2.2. **Domain-specific machinery loading (the leverage compound).** Every prompt loads the relevant context as prefix to the actual question. For our Ramanujan-flavor prompts, this meant:

- The 12-operator basis with one-line descriptions.
- Recent verified results (so the model knows what's already done).
- The toolkit-split prior (which operator clusters fire on which problem types).
- Specific catalogues to cross-check against (Berndt's notebooks [Ber05], Andrews–Berndt *Lost Notebook* [AB13], OEIS, arXiv).
- Recent gotchas from prior cycles (e.g., pre-declare low-complexity alpha-class to avoid tautological-fit).

Each new prompt incorporates what prior prompts learned. After 30 cycles, the prompts are loaded with accumulated calibration. The *hit rate climbs over time*—not because the model is smarter, but because the prompts are better-targeted.

2.3. **The verification-gate pattern (adversarial subagents).** This is the most important component. *Every numerical claim from the LLM proposer must pass independent verification.* We implemented this with adversarial subagents—distinct Claude / GPT-4 agent instances, each given the original LLM claim and asked to *actively try to break it*, not rubber-stamp it.

The subagent pattern in our run:

(1) LLM proposer produces a claim with numerical evidence at N digits.
(2) Independent subagent reimplements the construction from scratch.
(3) Subagent verifies at $N+$ digits across an expanded sweep of cells (orders of magnitude, edge cases, structural extensions).
(4) Subagent does explicit literature diff (local PDF grep + arXiv WebFetch + Google Scholar).
(5) Subagent returns: CONFIRMED / PARTIAL / REFUTED / DOWNGRADED.
(6) The verdict is the citable record, not the original LLM claim.

Three subagents in our run returned verdicts that *downgraded* an original LLM claim:

- **Level-pq Lambert annihilator family** (diary 55 → 58): math CONFIRMED at 25/25 cells × dps=200, but *novelty downgraded* from "infinite family of novel identities, 9.5/10, Trans. AMS, 6–12 pages" to "engine autonomously rediscovers classical $\dim E_2(\Gamma_0(pq)) = 3$ with elegant gcd parameterization, 4.5/10, Math. Mag. or Amer. Math. Monthly expository note, 2–4 pages" after the literature diff revealed the structure is classical.
- **PH spring-mass Bessel-image propagator speedup** (diary 53 → 59): math CONFIRMED at machine epsilon, but *speedup magnitude recalibrated* from "95× vs DFT" (which turned out to be vs mpmath-DPS DFT) to "1.7× vs numpy-FFT / 5,762× vs Verlet"—the same construction yields different speedup numbers against different baselines.
- 4 **cross-domain accelerators prior-art audit** (diary 60): all 4 implementations CONFIRMED, but 3 of 4 were attributed to prior literature (RetNet/Mamba-SSD for finite-pole attention, Crowdy–Marshall for annular Green, Schrödinger 1914 + Trovato 2016 for PH Bessel-image), with one (Airy Tricomi-U at $(1/6, 1/3)$ for $\mathrm{Ai}(-x)$) flagged as "DLMF-adjacent with parameters less-documented in canonical references."

One subagent CONFIRMED an original claim at high confidence:

- **AB Entry 4.2.3 defense** (diary 56 → 57): 5 independent verification paths converge (223/223 cells at dps=80; closed form at $\lambda = 1/2$, $\varepsilon = 1$ reduces both sides to $e^a \cdot \mathrm{erf}(\sqrt{a})$ exactly; independent Mittag-Leffler contour-integral proof via Bromwich/Hankel; literature audit found no published treatment since AB 2013—12 years unchallenged). *Recommended output: Ramanujan Journal corrigendum, 3–5 pages, with courtesy heads-up to Bruce Berndt before submission.*

This is the only result in the entire 30-cycle run that survives the verification gate as a *novel mathematical contribution*. Everything else is either a verified implementation of known math, a calibrated-honest negative, or a downgraded packaging contribution.

2.4. **Diary discipline + accumulating calibration.** Every cycle (productive or unproductive) produces a numbered diary entry under `diary/NN_topic.md`. The diary records:

- The original LLM claim.
- The verification verdict.
- The literature audit.
- Cross-links to prior cycles.
- Lessons captured (which get folded into subsequent prompts).

A parallel `DID_NOT_PAN_OUT.md` log records every refuted / corrected / downgraded speculation. In our run, this file has 19 entries—substantial enough to constitute a methodological contribution in its own right. *The credibility of the run depends on this file growing alongside the positive-finding log.*

The result is a workflow where every cycle's lessons compound. The hit rate climbs not because the model gets smarter but because the prompts and the verification gate get better-targeted.

3. Cycle-by-cycle taxonomy (30 cycles, May 2026)

We classify the ∼ 30 substantive ChatGPT-Pro cycles in our run as in table 1.

Category	**Count**	**Examples**
Novel mathematical finding (survived all gates)	1	AB 4.2.3 corrigendum
Verified implementation of known math	5	finite-pole attention, annular Green, PH Bessel-image, Airy Tricomi-U, CsCl Madelung
Engine-rediscovered classical structure (expository value)	1	Level-pq Lambert annihilator family
Calibrated-honest negative (model correctly says "no")	7	$\zeta(5)$ Apéry-like, XXX chain, Kepler equation, M3 P_{pq}, figure-8 three-body, Hubbard ladder, XXZ form factor
Confirmed but small / incremental	~ 8	various Bose–Einstein small-β / class-invariant G_n / etc
Cycle produced no actionable output	~ 8	various weaker prompts

TABLE 1. Cycle-by-cycle taxonomy of the ~ 30 substantive ChatGPT-Pro cycles in the Ramanujan-mining run, May 2026.

Of the ~ 30 cycles, the *single most consequential output* is the AB 4.2.3 corrigendum. That single result alone justifies the methodology—but only because the verification gate isolated it from the surrounding noise of plausible-looking-but-known or plausible-looking-but-wrong proposals.

The *calibrated-honest rate* is the most important methodological metric. *7 of ~ 30 cycles ($\sim$23%) returned a clean structural "no" with the obstruction identified.* This is the methodology's true-positive filter working—and it is the reason the surviving positive findings (the 1 novel + the 5 verified implementations) can be trusted.

4. The calibrated-honest cycles (representative)

We list the 7 calibrated-honest cycles, each with the obstruction the model correctly identified, in table 2.

In every case the model identified a *specific structural obstruction*—not a hand-wave "this is hard" but a precise reason why no closed form should exist. These are valuable as research direction signals (they tell us where NOT to spend more cycles) and as evidence that the model is calibrated.

5. What current LLMs are actually contributing

This is worth saying explicitly. In our run:

- *The math came from ChatGPT-Pro.* Derivations, closed forms, asymptotic analyses, structural arguments—all proposed by ChatGPT-Pro and (when correct) verified independently.
- *The methodology came from us.* Prompt design, machinery loading, verification orchestration, diary discipline, literature diff, prior-art attribution.
- *The verification subagents are also LLM-based.* Claude Opus 4.7 agents do the adversarial reimplementation. This is not circular—the proposer and the verifier are independent agent instances, each unaware of the other's reasoning trace.

Without ChatGPT-Pro, the methodology produces nothing. Without the methodology, ChatGPT-Pro produces a mix of correct, incorrect, and incidentally-novel proposals indistinguishably. *The combination is what filters reliably.*

Cycle	Domain	Obstruction identified
$\zeta(5)$ Apéry-like	Number theory	Koecher 1979 already gives a depth-2 expression; the pure single-summand depth-1 remains open.
XXX integrable chain (Lukyanov g^3)	Integrable physics	Closed forms exist for g^3 but g^4, g^5 require fitted parameters; not a derivation.
Kepler equation Padé–Lagrange	Celestial mechanics	Branch-point geometry forces slow Bessel–Kapteyn convergence; the Markley starter is essentially optimal in its class.
M3 P_{pq} Lambert series $\zeta(2k+1)$	Number theory	The known identities cover specific structural classes; the requested generalization is not derivable in the same framework.
Figure-8 three-body period	Celestial mechanics	Spectral hyperelliptic surrogate gives 30-digit precision but $P(\xi)$ is not algebraic/D-finite.
2-leg Hubbard ladder (isotropic)	Strongly correlated electrons	Lieb–Wu integrability does not extend; SO(8) Gross–Neveu IR mapping gives universal structure but not lattice e_0.
XXZ chain real-time correlator	Quantum dynamics	Caux–Konno–Sorrell–Weston [CKSW12] form factor has non-factorizing kernel $A_\xi(p_1, p_2)$ at interacting Δ; only $\Delta = 0$ (free fermion) factorizes.

TABLE 2. The 7 calibrated-honest cycles: each returned a precise structural obstruction, not a hand-wave.

6. What this implies for math-research workflow

We extract three implications:
(a) Frontier LLMs should be used as proposers, not as authorities. The proposer–verifier–gate–store pattern is robust; the "ask the LLM and trust the answer" pattern is not.
(b) Verification gates should be adversarial, not collaborative. A subagent told to "verify this claim" will rubber-stamp; one told to "actively try to break this claim, then report what survives" will catch real errors. In our run, three of four adversarial subagents returned downgrades—that's a 75% calibration miss rate in the proposer's original claims that would have shipped silently under a less-adversarial regime.
(c) The literature-diff step is non-optional. Three of four cross-domain accelerators in our run turned out to have direct prior publications. If we had skipped the prior-art audit, we would have shipped 4 "novel" claims that were actually 3 known + 1 partial. A computational tool paper that attributes its constructions correctly is still publishable; one that overclaims novelty is not.

7. Reproducibility

The full methodology—prompt templates, verification subagent prompts, diary entries, "did not pan out" log, paper drafts, code, per-cell CSVs—is publicly accessible at [Hel26].

The Erdős-style prompt template at `insights/CHATGPT_PRO_PROMPT_DESIGN_METHODOLOGY.md` is reusable for any subfield with three properties: (i) a small composable operator basis exists;

(ii) numerical verification is cheap (e.g., `mpmath` at 30+ digits is fast for the problem class); (iii) a literature corpus is locally accessible for prior-art audit.
Expected hit rate for a researcher with ChatGPT-Pro access running this workflow on a comparable corpus: ∼5–10% novel-math yield (1 of ∼ 30 cycles in our run), ∼50–70% useful-known-math yield, ∼20–30% calibrated-honest negative yield. The novel-math yield in particular depends heavily on whether the targeted subfield has any open questions that fall within the LLM's current capability envelope.

8. Limitations

The 1-in-30 novel-math rate may not generalize. Our subfield (Ramanujan-style q-series, theta functions, Lambert identities) has a substantial open-question landscape that maps well to current LLM capabilities. Other subfields may have higher or lower yield.
The verification subagents are themselves LLM-based and could share blind spots with the proposer. We mitigate this with explicit numerical bars (`mpmath` at 30+ digits is hard to fake) and multiple independent verification paths (in the AB 4.2.3 case: 5 distinct paths, including a `sympy`-symbolic reduction that is a non-neural component).
The literature-diff step is incomplete. WebFetch + arXiv + Google Scholar miss paywalled / non-indexed / older / non-English material. For our run, this caught only the obvious prior-art cases; deeper checks against textbook material (e.g., Olver's *Asymptotics and Special Functions* for Airy Tricomi-U) require offline access we did not have.
The "did not pan out" log is a credibility floor, not a ceiling. A research run that publishes its 19 refuted/downgraded speculations alongside its 1 surviving novel finding is more trustworthy than one that publishes only the wins. But the trustworthiness comes from external scrutiny of the log, not from the log itself.

9. Conclusion

We have documented a four-component workflow—Erdős-style prompts, machinery loading, adversarial verification subagents, diary discipline—that filters LLM mathematical proposals reliably enough to produce publication-grade output. In a 30-cycle private research run, the workflow produced 1 novel mathematical finding (an AB *Lost Notebook* corrigendum), 5 verified computational primitives (cross-domain accelerators), 1 elegant rediscovery of classical structure, and 7 calibrated-honest negatives (where the model correctly identified the obstruction). The single most consequential output is the corrigendum; the most important methodological finding is the 75% downgrade rate of the adversarial verification subagents, which says that the unverified proposer rate of overclaim is high enough that the verification gate is essential.

We release the full workflow at [Hel26] for reproduction and critique.

Acknowledgments

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